Applied NLP

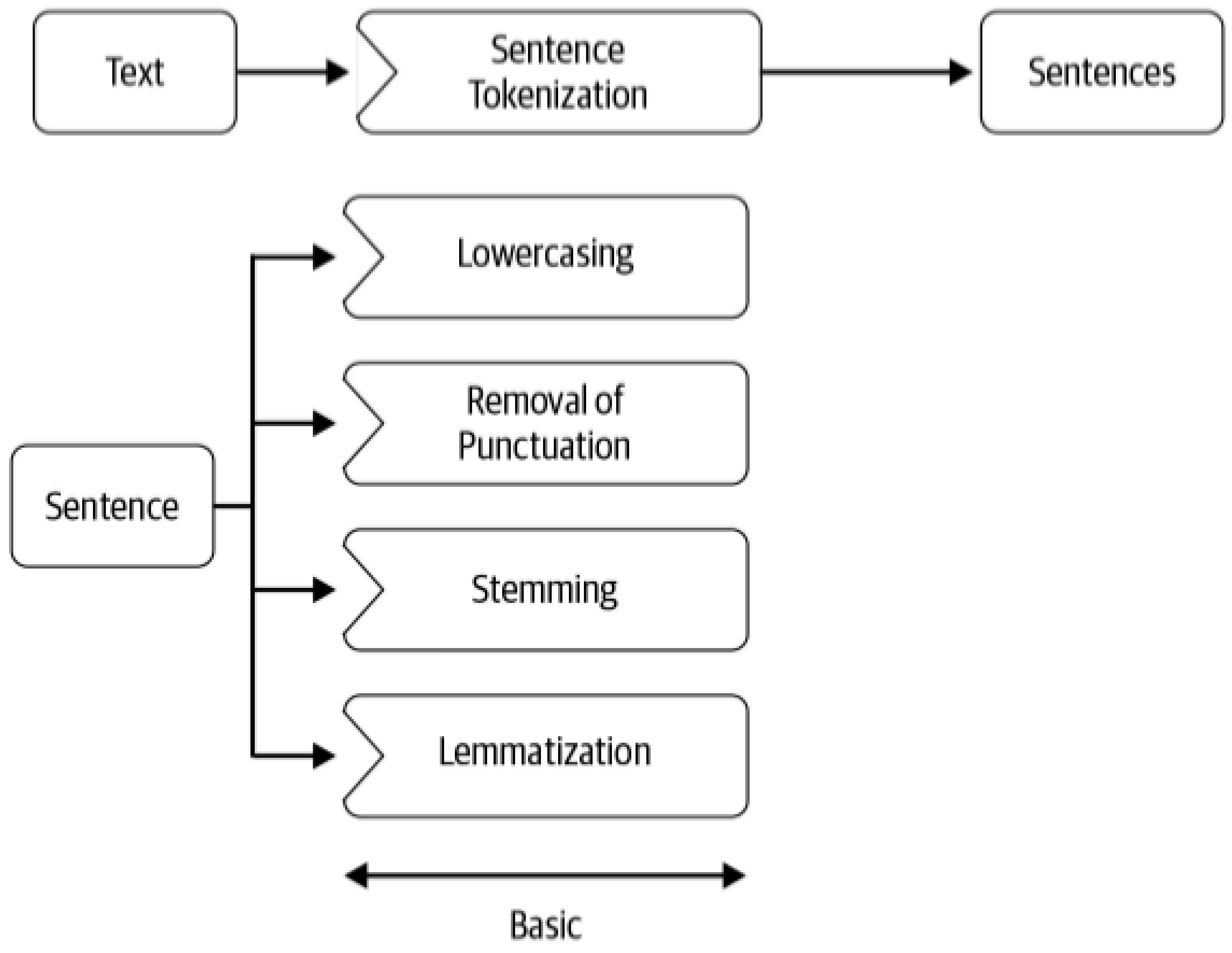
DATA 641

Lecture 2 — NLP Pipeline; Pre-processing

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Pre-processing

Some common Pre-Processing Steps:



Word tokenization--Types and Tokens

- Type = abstract descriptive concept
- Token = instantiation of a type

e.g. `To be or not to be`

Here we have 6 tokens (`to`, `be`, `or`, `not`, `to`, `be`) and 4 types (`to`, `be`, `or`, `not`)

Types = the vocabulary; the unique tokens

Whitespace

Whitespace tokenization should be old-hat by now: `text.split(" ")`

"Every time I learn something new, it pushes some old stuff out of my brain."

`["Every", "time", "I", "learn", "something", "new,", "it", "pushes", "some", "old", "stuff", "out", "of", "my", "brain."]`

Punctuations

We typically don’t want to just strip all punctuation, however

1. Punctuation signals boundaries (sentence, clausal boundaries, parentheticals, asides)
2. Some punctuation has illocutionary force, like exclamation points (!) and question marks (?)
3. Emoticons are strong signals of e.g. sentiment

Stemming and Lemmatization

Many languages have some inflectional and derivational morphology, where similar words have similar forms:

- organizes, organized, organizing

Stemming and lemmatization reduce this variety to a single common base form

Stemming

Heuristic process for chopping off the inflected suffixes of a word:

e.g.

organizes, organized, organizing --> organ

Lemmatization

Using morphological analysis to return the dictionary form of a word (the entry in a dictionary you'd find all forms under)

e.g.

organizes, organized, organizing --> organize

Text normalization: Some common steps for text normalization are to convert all text to lowercase or uppercase, convert digits to text (e.g., 9 to nine), expand abbreviations, and so on.

Language detection: Since a majority of the pipeline is built with language specific tools, what will happen to our NLP pipeline, which is expecting English text? In such cases, language detection is performed as the first step in an NLP pipeline.

Code mixing and transliteration: Many people across the world speak more than one language in their day-to-day lives. Thus, it's not uncommon to see them using multiple languages in their social media posts, and a single post may contain many languages.